CMPE 2965 - Technical Project
Work Breakdown Structure
Group 8 - Sadie, Lovelyn

Noise Pollution Monitoring and Reporting Device **(100)**

1. Project Management **(8)**
    - 1.1. Proposal, specification, and verification tests **(5)**
    - 1.2. Github project **(3)**
2. Hardware **(20)**
    - 2.1. Component selection and and procurement **(3)**
    - 2.2. Circuit design **(5)**
    - 2.3. Breadboard prototype **(5)**
    - 2.4. Enclosure design and 3D printing **(3)**
    - 2.5 . Hardware integration and bench testing **(4)**
3. Embedded Firmware **(22)**
    - 3.1. Sound sampling (mic reading, dB conversion per ISO 1996-2:2017 methodology) **(6)**
    - 3.2. GNSS integration **(4)**
    - 3.3. Cellular communication **(4)**
    - 3.4. Power / recording, switch handling **(2)**
    - 3.5. Status LED logic **(2)**
    - 3.6. Bluetooth communication with mobile application **(4)**
4. Web Server **(20)**
    - 4.1. Front end web app **(11)**
        - 4.1.1. Allows user to select a recording session to view **(3)**
        - 4.1.2. Displays raw data for user and allows it to be downloaded **(4)**
        - 4.1.3. Processes and interprets data (eg graphs) for the user **(4)**
    - 4.2. Data handling server **(9)**
        - 4.2.1. Receives and organizes data from the device into files to be stored in the database **(4)**
        - 4.2.2. Provides data to the web app about various recording sessions **(3)**
        - 4.2.3. Provides recording session data to web app **(2)**
5. Mobile Application **(10)**
    - 5.1. Monitors device and provides diagnostic information **(6)**

- 5.2. Allows certain settings to be manually set (ex location and time when no GPS) **(4)**

6. Integration and Testing **(13)**
   - 6.1. Device-to-server communication testing **(3)**
   - 6.2. Full system integration **(5)**
   - 6.3. Verification testing against all specs **(3)**
   - 6.4. Field testing ? **(2)**
7. Documentation **(7)**
   - 7.1. User Manual **(2)**
   - 7.2. Final Technical Report **(3)**
   - 7.3. Final Presentation Materials **(2)**